Lab 0

Prerequisite

Challenge 1

shell 命令的用法介绍与实操

- `touch`
  - 用法：修改文件时间戳，若目标文件不存在且未提供 `-c` 或 `-h` 则创建一个空文件。
  - 实操：

```
zzh@LAPTOP-K6JHIFVL:~/test$ ls -l
total 0
-rw-r--r-- 1 zzh zzh 0 May 31 01:33 example
zzh@LAPTOP-K6JHIFVL:~/test$ touch example
zzh@LAPTOP-K6JHIFVL:~/test$ ls -l
total 0
-rw-r--r-- 1 zzh zzh 0 May 31 01:44 example
zzh@LAPTOP-K6JHIFVL:~/test$ touch example1
zzh@LAPTOP-K6JHIFVL:~/test$ ls -l
total 0
-rw-r--r-- 1 zzh zzh 0 May 31 01:44 example
-rw-r--r-- 1 zzh zzh 0 May 31 01:44 example1
```

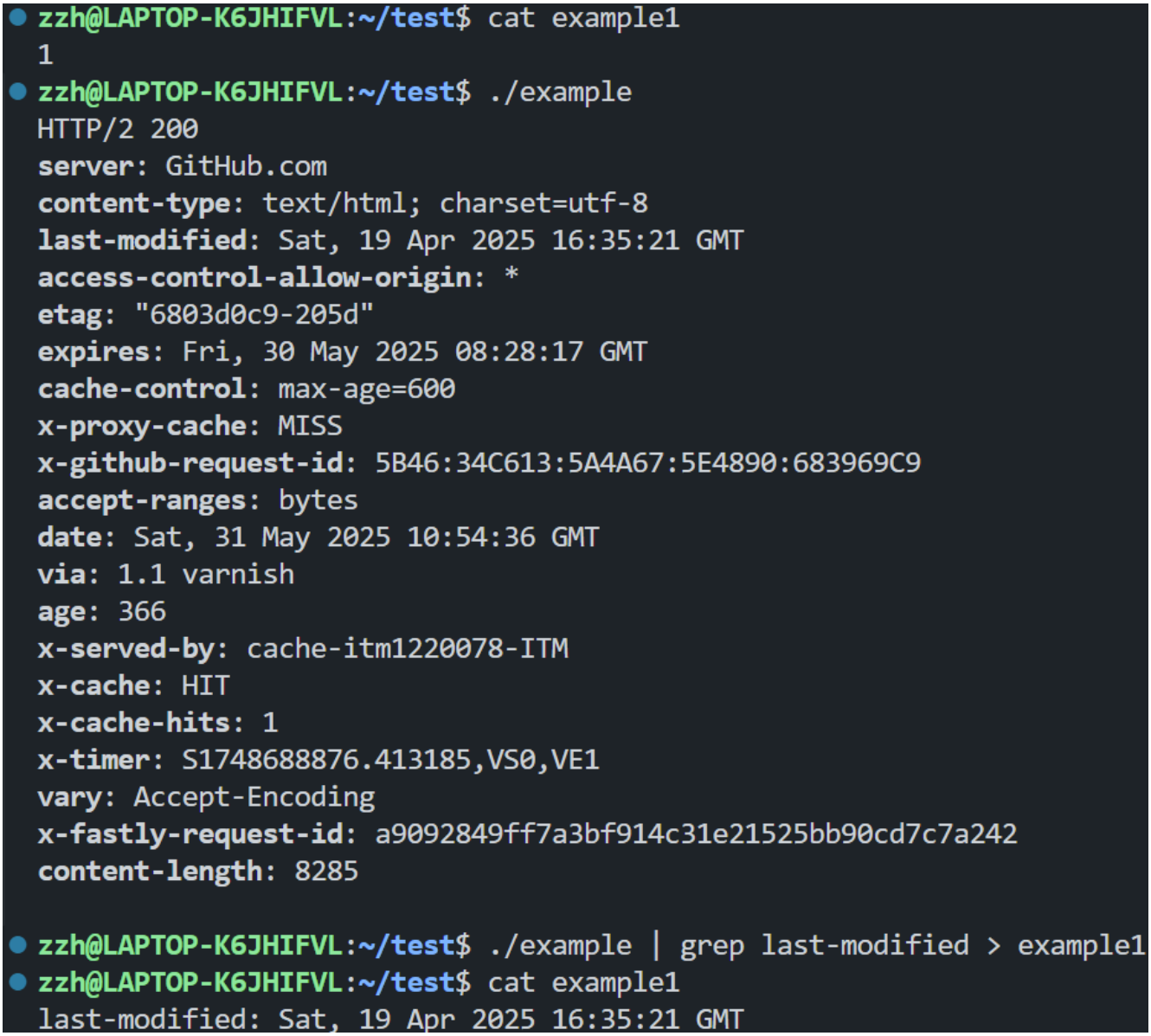
- `cat`
  - 用法：连接文件并打印到标准输出。
  - 实操：

```
zzh@LAPTOP-K6JHIFVL:~/test$ cat example
#!/bin/sh
curl --head --silent https://missing.csail.mit.edu
zzh@LAPTOP-K6JHIFVL:~/test$ cat example1
1
zzh@LAPTOP-K6JHIFVL:~/test$ cat example example1
#!/bin/sh
curl --head --silent https://missing.csail.mit.edu
1
```

- chmod
  - 用法：更改文件或目录的权限。
  - 实操：

```
zzh@LAPTOP-K6JHIFVL:~/test$ ls -l
total 8
-rw-r--r-- 1 zzh zzh 7 May 31 18:21 example
-rw-r--r-- 1 zzh zzh 2 May 31 18:03 example1
zzh@LAPTOP-K6JHIFVL:~/test$ cat example
echo a
zzh@LAPTOP-K6JHIFVL:~/test$ ./example
bash: ./example: Permission denied
zzh@LAPTOP-K6JHIFVL:~/test$ chmod +x example
zzh@LAPTOP-K6JHIFVL:~/test$ ./example
a
```

- grep
  - 用法：搜索并打印文本文件中符合条件的行。
  - 实操：



Saint John

搜索可知 `fuser` 命令可以用来查找正在使用某个文件的进程， `kill` 掉对应进程即可。

```
admin@ip-10-1-13-94:~$ fuser /var/log/bad.log
/var/log/bad.log:       593
admin@ip-10-1-13-94:~$ kill 593
admin@ip-10-1-13-94:~$ tail /var/log/bad.log
2025-05-31 11:10:38.869452 token: 1006543806
2025-05-31 11:10:39.169863 token: 1706795253
2025-05-31 11:10:39.470281 token: 1898470008
2025-05-31 11:10:39.770691 token: 1864968491
2025-05-31 11:10:40.071084 token: 1090255845
2025-05-31 11:10:40.371490 token: 1823113750
2025-05-31 11:10:40.671907 token: 117313606
2025-05-31 11:10:40.972318 token: 992633147
2025-05-31 11:10:41.272738 token: 1276786828
2025-05-31 11:10:41.573161 token: 1937129385
admin@ip-10-1-13-94:~$ tail /var/log/bad.log
2025-05-31 11:10:38.869452 token: 1006543806
2025-05-31 11:10:39.169863 token: 1706795253
2025-05-31 11:10:39.470281 token: 1898470008
2025-05-31 11:10:39.770691 token: 1864968491
2025-05-31 11:10:40.071084 token: 1090255845
2025-05-31 11:10:40.371490 token: 1823113750
2025-05-31 11:10:40.671907 token: 117313606
2025-05-31 11:10:40.972318 token: 992633147
2025-05-31 11:10:41.272738 token: 1276786828
2025-05-31 11:10:41.573161 token: 1937129385
```

Check My Solution

Solution is correct, you made a sad server happy, congrats!

It took you 2 minutes and 11 seconds.

You used 0 clues.

Your earned 1 points.

Challenge 2

python 程序执行与功能解释

```
(base) zzh@LAPTOP-K6JHIFVL:~/coding/python$ ./stringconverter
give me your string: asdGHJ456:"
length of string: 11
now your string: ASDghj456:"
```

功能：接受任意字符串，输出字符串长度并将字符串中小写英语字母转换为大写。

calculator

```python
from pwn import *
import re

def solve_challenge(host='10.214.160.13', port=11002):
    # 使用pwntools创建连接
    conn = remote(host, port)

    # 连续解决10个挑战
    for i in range(10):
        expr_line = None
        while not expr_line:
            # 接收数据直到新行
            data = conn.recv().decode()
            print(data.strip())
            # 使用相同的正则表达式提取数学表达式
            match = re.search(r'([0-9+\-\s]+)=', data)
            if match:
                expr_line = match.group(1).strip()

        # 计算答案
        answer = str(eval(expr_line)) + '\n'
        print(answer)

        # 发送答案
        conn.send(answer.encode())

    # 接收最终结果
    result = conn.recv()
    print(result.decode())

    # 关闭连接
    conn.close()

if __name__ == '__main__':
    solve_challenge()
```

```
zzh@LAPTOP-K6JHIFVL:~$ /home/zzh/miniforge3/envs/ctf/bin/python /home/zzh/coding/python/calculator.py
[+] Opening connection to 10.214.160.13 on port 11002: Done
================================================
Mom: finish these 10 super simple calculations,
     and you will get a flag
Melody: that's easy...
Mom: yep, in 10 seconds
================================================
2040324543 + 517425748 - 1557184245 + 74520728 - 1693976112 - 1701878628 + 58639069 + 879033355 + 688890384 + 388835381 =
-305369777

Good, next:
281768984 + 1940011378 - 320485095 - 1369765990 - 1106920342 + 1661582079 + 899176944 + 125734590 + 1300699005 - 863839295 =
2547962258

Good, next:
1530143214 + 447221202 + 1706738859 - 1698491622 + 730036347 - 1427207922 + 342058433 + 805338307 - 1820911597 + 382041296 =
996966517

Good, next:
76411622 + 1780109529 - 2075147555 + 1284466107 - 2106099437 - 2075554381 + 419056030 + 930057638 + 1701546493 + 226456836 =
161302882

Good, next:
420908969 - 1295904523 - 725916763 + 1425510931 + 2128863480 + 1179693004 + 1703691955 + 1033380021 - 570679356 + 712235126 =
6011782844

Good, next:
171184235 + 2018054148 + 1015597794 + 568551506 - 1281693752 - 853759832 + 301189453 - 318086475 - 2103423539 - 1426450929 =
-1908837391

Good, next:
1403665986 - 2142594826 + 1270979171 - 443746507 - 1198853681 + 223470025 - 1796287492 - 1960278182 - 894668665 - 1234062278 =
-6772376449

Good, next:
1176354383 + 1731481490 - 2090002819 + 357166404 - 1645681724 - 1983262433 - 94793972 - 1861738355 + 859381164 - 1059948023 =
-4611043885

Good, next:
1313982847 + 1280789113 + 1377458575 + 1627188329 - 1474929688 - 153522361 + 654105423 + 1981344259 - 1787088168 + 495083656 =
5314411985

Good, next:
1974644349 + 39666658 + 161752405 + 1287727817 + 2059406227 - 931884766 - 1483610241 - 41208469 - 1158773879 + 838228760 =
2745948861


Good, next:

Good job, here is your flag: AAA{melody_loves_doing_calculus_qq_qun_386796080}

[*] Closed connection to 10.214.160.13 port 11002
```

flag: `AAA{melody_loves_doing_calculus_qq_qun_386796080}`

Misc

Challenge 1

将所给字符串

```
;cR!f<`MjL9eg<R<bl5:;D9go<,6#?<CUXo<)lXK;cPZ";D9h.=);&:<bu;I9l*Er<A8VO<GYQ"<blqZ<%p0t<bbf5<GY,q
```

输入 cyberchef,经由自动识别依次进行 `From Base85` 、 `From Base64` 、 `From Base32` 解码，得到 flag:

`AAA{wELc0m3_t0_Ctf_5umMEr_c0UrsE_2025}`

Challenge 2

flag 第一部分：利用 StegSovlve 查看 R/G/B 通道的最低位平面均可发现文字，
为 `AAA{gr3@t_J08!_1et'5_`



flag 第二部分：利用 `strings` 命令，观察到文件尾异常字节，为 `P1@y_m1SC_TOG3Th3R}`
拼接为 flag: `AAA{gr3@t_J08!_1et'5_P1@y_m1SC_TOG3Th3R}`